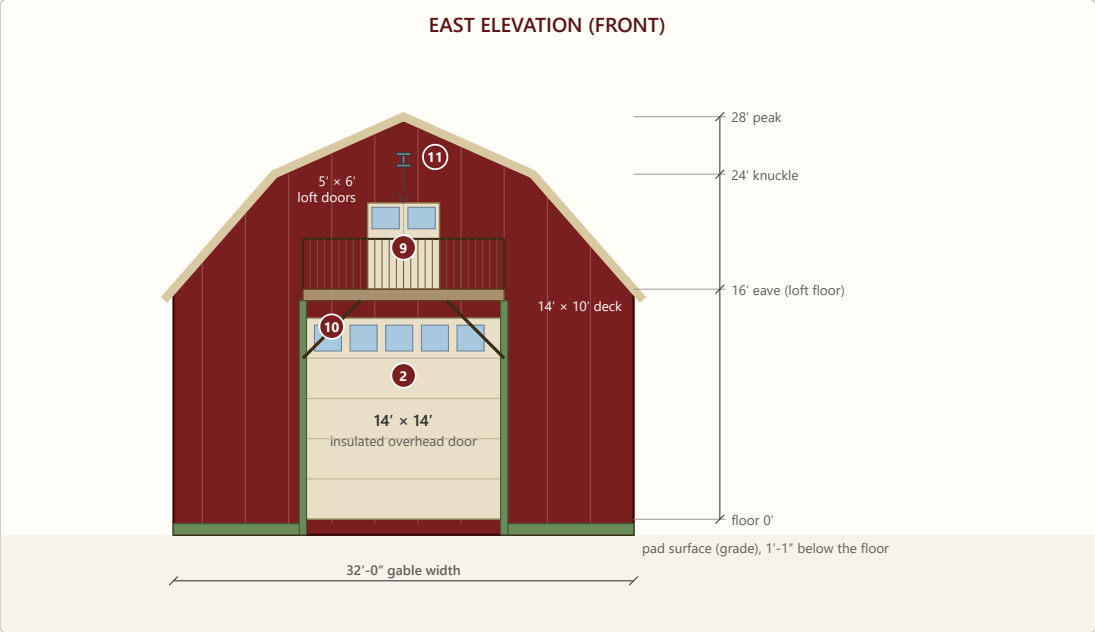
Workshop Pole Barn

Mountain Prairie Acres · 32 × 48 ft gambrel pole barn: RV garage, workshop, and storage loft · **PLANNED FOR 2028** · owner-build

Every drawing, count, and price in this package is computed from one design model at the planned footprint in the site data. County rules sets the building against what Elmore County's ordinance says.

> **Planned for 2028. Nothing is ordered, permitted, or built.** These are planning drawings and figures. Have Elmore County and an Idaho-licensed engineer confirm every load and dimension, and get quotes, before anything is ordered. Prices are 2026 ballpark figures.

The Building in Brief

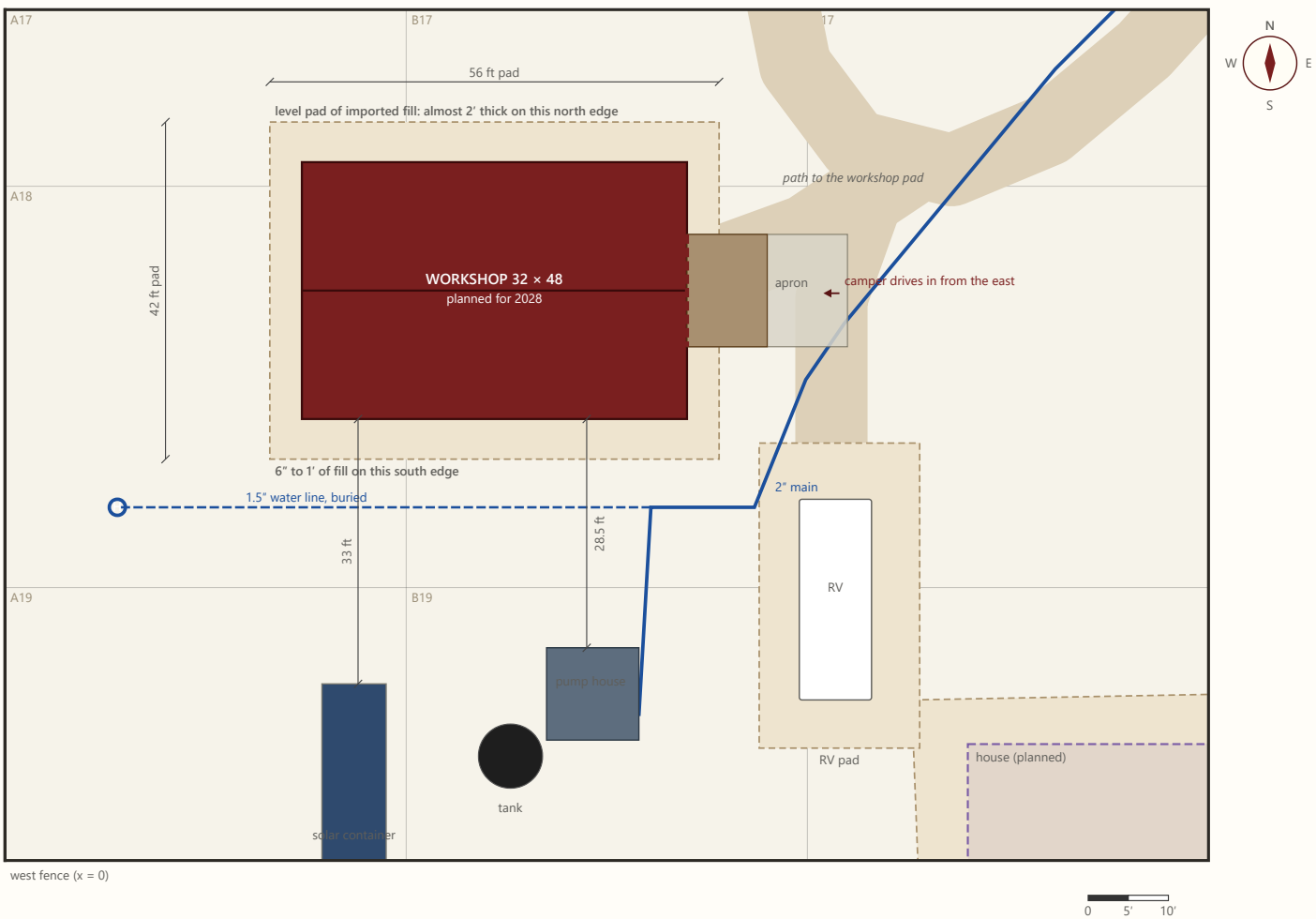


East elevation, the front: the 32′ × 48′ building seen from the driveway side. Every picture in the package is drawn from one design model, so the length lives in one place.

Use	RV garage, workshop, and storage loft. What the building really holds decides its permit path: see County rules
Footprint	32 × 48 ft: 1,536 sq ft on the main floor and the same on the loft floor, of which about 950 sq ft has 7 ft or more of headroom (19.75 ft wide down the middle)
Bays	Sidewall posts at 8 ft: 8-8-8-8-8-8, with 19 piers and 25 trusses at 2 ft OC. The length is a whole number of 8 ft bays, so the posts, the 16 ft girts, the 4×8 sheets, and the 36 in steel panels land on a clean module
Heights	16 ft eave (the loft floor), 24 ft at the pitch break, 28 ft at the ridge, all above the floor. The floor is 1′-1″ above the pad, so the ridge is 29′-1″ above the pad and the mid-point of the gable 23′-1″. The gambrel slopes are 49° below the break and 24° above it
Orientation	Long side east–west; the main door is on the **east** gable, facing the paths and the RV pad, so a prevailing west wind meets the closed back wall. The baselines do not settle which way the winter wind blows
Design loads	120 psf ground snow (the county's zone; confirm) · 115 mph wind, Exposure C · 40+ psf loft floor · seismic category set by the engineer
Main door	14′ × 14′ insulated, wind-rated sectional overhead door, centered on the east gable
Loft doors	5′ × 6′ insulated double doors on the east gable, above the main door at loft-floor level, opening onto the deck
Hoist beam	Cantilevered W6 steel I-beam, 3 ft out over the deck, for a 500–1,000 lb chain fall
Landing deck	14′ × 10′, free-standing on 6×6 posts to grade, 42″ rail; reached from the loft only (no outside stairs)
Walk door	3′ × 7′ insulated steel, south wall near the east end
Windows	3 in the north wall and 3 in the south wall, 4′ × 3′ insulated sliders with the sill 5′ above the floor, centered in their bays and clear of the stair and the walk door; one 3′ × 4′ single-hung in the west gable at the loft. About $1,900 to $3,500 in materials with their flashing and trim
Loft access	L-shaped stairs: first flight along the south wall, a landing at the southwest corner, second flight up the west wall
Heat and power	Wood stove and a mini-split heat pump (cooling too) · 200 A sub-panel; where it is fed from is still open
Colors	beige roof and trim · barn-red walls
Build approach	Owner-build with pre-fab stamped trusses, in phases if needed. The goal is weather-tight before snow flies
Budget	Materials about $80,900 to $125,100; about $105,500 to $179,500 all in, with tax, permits, engineering, utilities, rentals, and a 15% contingency (materials and cost)

Where It Sits

The workshop goes on the graded pad north of the pump house and the containers, with its east door lined up with the road-mix path and the RV path. Its footprint is in `data/site.json` at 32 × 48 ft, centered on the pad.



The pad, the planned building, and what is around it, drawn from `data/site.json` on the 50 ft master grid. Distances are measured on the grid. Compare the homestead plot map.

Planned footprint	48 × 32 ft in `data/site.json` (feature `workshop`), long side east–west, the same 48 × 32 ft this package is drawn at
Center	**B18-E11-S13** (cell B18), x 61, y 863 · 43.473085, -115.594602
Corners	northwest A17-E37-S47 · northeast B17-E35-S47 · southwest A18-E37-S29 · southeast B18-E35-S29
Grid cells it touches	A17, B17, A18, B18
Pad	56 × 42 ft (`workshop_pad`, edges approximate), level. The building is centered on it, leaving 4 ft of pad on the west and east, and 5 ft on the north and south.
Fill on the pad	Imported fill, from the owner (October 2026): **almost 2 ft over the original ground along the north edge, 6 in to 1 ft along the south edge.**
Original ground under the footprint	The lidar, taken before the pad was filled, puts it between 4,757.6 and 4,759.1 ft: northwest 4,757.6, northeast 4,758.3, southwest 4,758.1, southeast 4,759.1. It rises toward the south and east, which fits the thicker fill on the north.
Pad surface (estimate)	About 4,759.7 ft: the original ground at the center (4,758.4) plus about 1.3 ft of fill. Shoot it with a level before anything is poured.
Neighbors	Pump house 28.5 ft south of the south wall · solar container 33.0 ft south · buried 1.5 in water line 11.0 ft south · water line along the west fence 16.4 ft west · RV pad 9.5 ft to the southeast · irrigation tank 38.0 ft south · planned house 53.5 ft to the southeast

Notes on the pad and the foundation

- **Fill and compaction.** Ask the engineer whether the fill was placed in compacted lifts. If that is not documented, plan to compact and test it, or to deepen the gravel base, before the slab goes on it.

- **The pier holes go through the fill.** The 60 in is measured from the pad surface, so the north row's holes pass through about 2 ft of fill and end about 3 ft into native soil, and the south row's end about 4 to 4½ ft into it. Each pier pad should bear on undisturbed native soil, or on fill the engineer accepts. Detail 3 and Section A draw this.
- **The north edge.** The fill is thickest along the north edge, where the pad reaches only 5 ft beyond the wall (4 ft past each end). Check that the edge is firm and not slumping, and keep the north wall's piers and the slab edge well inside it.
- **The pad surface.** The lidar predates the pad, so shoot the finished surface with a level and record it here. The slab, foam, and gravel stack up about 13 in above it, so the floor sits about 1′-1″ above the pad.

County Rules and What They Mean Here

From Elmore County's published Zoning and Development Ordinance (Title 7, chapters 1 and 2, adopted May 2018, amended 2023 and 2024), the building regulations (Title 9), and the county's Notice of Proposed New Agricultural Building, read in October 2026 at elmorecounty.org. The ordinance is being amended (case ZOA-2026-01, to match the updated Comprehensive Plan; the staff report for the October 22, 2026 hearing recommends continuing the review to November 18, 2026). The draft keeps the height table and the accessory- and agricultural-structure standards cited below, but it reorganizes chapter 2, so section numbers may change. This is a planning summary, not legal advice and not the county's answer: read the current text and get each answer in writing.

Rule	What the ordinance says	This building	Where it stands
Height limit of the zone Table 7-2-27 (A)	35 ft in the Ag, Rec, and RR districts. An agricultural structure in the Ag district may go to 80 ft.	Ridge 29′-1″ above the pad; mid-gable 23′-1″.	Within the limit. Confirm which district the parcel is in.
Height of an accessory building 7-2-42 D, and "building height" in Title 7 chapter 1	No taller than the house or 24 ft, whichever is greater. In a residential district, a building in the front yard may not be taller than the house at all. Height runs from the average ground at the front wall to the average height (mid-point) of the highest gable.	Mid-gable 23′-1″ above the pad surface, or 24′-5″ above the original ground (the fill averages 1.3 ft under the east wall).	Close: under by 11″ measured from the pad, over by 5″ from the original ground, if the house is no taller than 24 ft. Averaging the gambrel's whole gable, not just halving eave and ridge, gives about 24.5 ft from the pad. Ask how the county reads a gambrel and where it takes grade.
Size of an accessory building 7-2-42 C	In the front yard, no more than 1,500 sq ft or half the house's footprint, whichever is less.	1,536 sq ft. The planned 40 × 66 ft house is 2,640 sq ft, so half is 1,320 sq ft.	Over the cap, if the building counts as in the front yard (between the road and the front of the house). Ask which road is the front. At 32 ft wide, a length of 46 ft (1,472 sq ft) would clear the 1,500 sq ft figure, and 41 ft (1,312 sq ft) would clear half the house too.
A house comes first 7-2-42 A	An accessory building needs a house on the same parcel with a certificate of occupancy or a valid building permit. It may not be a dwelling or be used for business.	The house is still planned.	Permit the house first, or be ready to show it under a valid permit.
Matching the house 7-2-42 E	Roofing and finish similar in color to the house, and a roof pitch similar to the house's. In a residential district, a front-yard building of 1,500 sq ft or more must also portray the house's architectural character.	Barn-red walls, a beige roof, and a gambrel (49° and 24°). At 1,536 sq ft it is over the size where the character rule starts.	Likely to be questioned. Ask how strictly the county applies it, or choose colors and a pitch that suit the house.
Agricultural building Title 9, 9-1-7; 7-2-46; the definitions in Title 7 chapter 1; the county's exemption form	Exempt from the building code and its fees, but not from zoning: it needs zoning approval first and the county's agricultural exemption, and the district's setbacks and height still apply. It must stand on a farm (5 acres or more in agricultural use) and be used **solely** to store farm implements, hay, grain, poultry, livestock, or horticultural products; the definition excludes personal items and a shop not related to agriculture. The owner signs a notarized statement, recorded with the county, and the county may inspect within 12 months to confirm the use is still agricultural.	Drawn as an RV garage, a workshop, and a storage loft.	Does not fit as drawn: an RV and a general workshop are personal use. A barn for orchard equipment, a tractor, bins, and harvest storage would fit if the parcel counts as a farm, and then the plans and the signed statement must describe that use. A later change of use may be a violation.
A 14 ft wall and a 27 ft peak not found	Not in the published ordinance (Title 7 chapters 1 and 2, Title 9, or the agricultural-building form).	Eave 16 ft and ridge 28 ft above the floor.	Unverified. If the county confirms them, ask for the section in writing. A 14 ft wall would put the loft floor at 14 ft, so the 14 ft door and the 11′-10″ camper would not fit under it: that is a redesign, not an adjustment.

Which permit path is this building on?

The county sorts a building by what it is used for, not by its name.

- **Accessory building: the design as drawn.** An RV garage, a workshop, and a storage loft. The house comes first, and the height, size, and matching rules above apply. It needs a building permit, plan review, and inspections.
- **Agricultural building.** A barn for orchard equipment, a tractor, bins, and harvest storage. It is exempt from the building code, but not from zoning: it needs zoning approval before construction and the county's agricultural exemption in place of a building permit, and the district's setbacks and height still apply. It must stand on a farm (5 acres or more in agricultural use, tied to the assessor's tax categories), so ask whether the parcel qualifies. The owner signs a notarized statement, recorded with the county, that the building is used solely for agriculture.

> The two cannot be mixed: the definition says *solely*. With the RV inside, the building is on the first path, and a building later put to another use can be required to obtain every permit and prove it meets the code. If the plan changes to a farm barn, the package should be redrawn for that use (a different name, no camper clearances), and the application should describe what the building will hold.

Before Any Site Work

1. **Call the Elmore County Land Use and Building Department** (208-587-2130, extension 6500): confirm the ground snow load, wind speed, frost depth, and exposure for this parcel, and which rules apply at about 4,750 ft.
2. **Settle the permit path with the county, in writing** (see County rules): the parcel's zoning district (some of the rules apply only in residential districts); whether this is an accessory building or an agricultural building, which follows from what it will really hold; how the height is measured, from what grade, and how a gambrel is averaged; the front-yard size cap; and whether the house must be permitted first.
3. **Get stamped truss drawings** from a pre-fab truss company, with the loft stair opening, the loft doors, and the hoist-beam loads on them.
4. **Get stamped foundation and structural plans** from an Idaho-licensed engineer, including what to do about the pad's fill.
5. **Pull the building permit.**
6. **Call 811** before any digging. The buried 1.5 in water line is about 11 ft south of the south wall and the west-fence line about 16 ft west.
7. **Settle the power source** and run the sub-feed conduit before backfill.
8. **Shoot the pad's elevation** with a level and get the fill checked before the footings.

Open Questions

1. **What will the building really hold?** That decides the permit path (see County rules). An RV garage and a workshop make it an accessory building, which is what these drawings show. A barn for orchard equipment and harvest could be an agricultural building. It cannot be both.
2. **Which snow load applies?** The county's 120 psf zone is used here; the permit page ties it to buildings above about 5,000 ft.
3. **Where does the power come from?** The pump house with its solar inverters and batteries, or the house? A 200 A panel is a panel rating, not a supply: check that the source can run the welder (50 A), the compressor (30 A), and the mini-split together.
4. **What is the hoist for?** Lifting from the deck needs only the 3 ft beam. Lifting from the ground has to clear the 10 ft deck, so the beam would reach about 12 ft out and the engineer would size it and its back-span.
5. **How far does the slab slope?** The slab slopes toward the east door. At ¼ in per ft over 48 ft that is 12 in, which is a lot. Slope only the 12 to 16 ft nearest the door, or use ⅛ in per ft; decide with the concrete subcontractor.
6. **Stove, condenser, and panel.** The wood stove's spot is not chosen. The mini-split condenser goes on the west or the south wall.
7. **The winter wind.** The door is on the east end on the idea that a prevailing west wind hits the closed back wall. If the winter wind comes from the east instead, the door and the deck take it and the drifts pile up there. The weather station's first year should settle it.
8. **Camper height.** The drawings use an 11′-10″ camper roof, which leaves 2′-2″ under the door head and 3′-4″ under the loft. Measure yours.

How to Use This Package

1. Verify the design loads and the county rules (see County rules) with Elmore County before any orders go out.
2. Use the specs when calling the truss plant or the steel vendor; read straight off the page.
3. Use the plans and elevations for layout on site and with the permit.
4. Use the sections and details when framing, sheathing, and siding; the numbered and lettered markers match the legends.
5. Use the materials list as the shopping and check-off list. Firm prices replace the estimates as quotes come in.

Sources

- Site records: site master grid and homestead plot map (`data/site.json`: the planned workshop, its pad, the paths, the water lines, and the neighbors), and the USGS lidar terrain.
- County building values: site baselines (Elmore County Land Use and Building Department: snow zone, frost depth, wind, and seismic zone).
- County zoning and building rules: Elmore County Zoning and Development Ordinance, Title 7 chapters 1 and 2 (definitions; table 7-2-27; sections 7-2-42 and 7-2-46), Title 9 section 9-1-7, the Notice of Proposed New Agricultural Building with its exemption statement (elmorecounty.org, October 2026), and the staff report and draft text for ZOA-2026-01 (October 22, 2026 hearing).
- The pad's fill, level, and thickness: the owner, October 23, 2026.

Mountain Prairie Acres · Workshop pole barn overview · updated October 23, 2026 · every figure is drawn from `docs/workshop_model.js`, and `tools/test_workshop_model.js` checks it against `data/site.json` and the county values in `data/site_baselines.json`.